

- 6 (a) At the end of glycolysis in mammalian muscle tissue, pyruvate is formed. Pyruvate can be further metabolised along two different pathways, pathway 1 and pathway 2.

Fig. 6.1 outlines the two pathways.

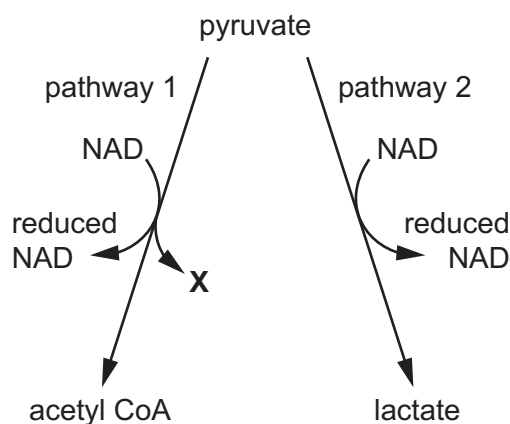


Fig. 6.1

- (i) State the condition under which pyruvate would be metabolised along pathway 2.
 [1]
- (ii) Name compound **X**.
 [1]
- (iii) Name the two **types** of reaction that are involved in pathway 1.

 [2]
- (iv) Suggest a cell in a mammal that **cannot** carry out pathway 1.
 Give a reason for your answer.

 [2]
- (v) ATP is synthesised during glycolysis.
 Name the process by which ATP is synthesised.
 [1]

- (b) The rufous hummingbird, *Selasphorus rufus*, feeds on the nectar produced by flowers. Nectar is a liquid containing sugars. Hummingbirds hover to maintain a position next to a flower while they feed. This requires a lot of energy.

Fig. 6.2 is a rufous hummingbird.



Fig. 6.2

Hummingbirds that have not fed for some time have a respiratory quotient (RQ) value of 0.7. After feeding their RQ value is 1.0.

- (i) Define the term *respiratory quotient*.

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..... [2]

- (ii) Explain why a hummingbird that has not fed for some time has an RQ value of 0.7, while a hummingbird that has fed has an RQ value of 1.0.

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..... [2]

- (iii) When animals exercise they can respire substrates obtained directly from their diet, as well as respiring substrates from energy stores.

An investigation worked out the percentage of respiration that used substrates obtained directly from the diet of humans and hummingbirds as they exercised.

Fig. 6.3 shows the results of this investigation.

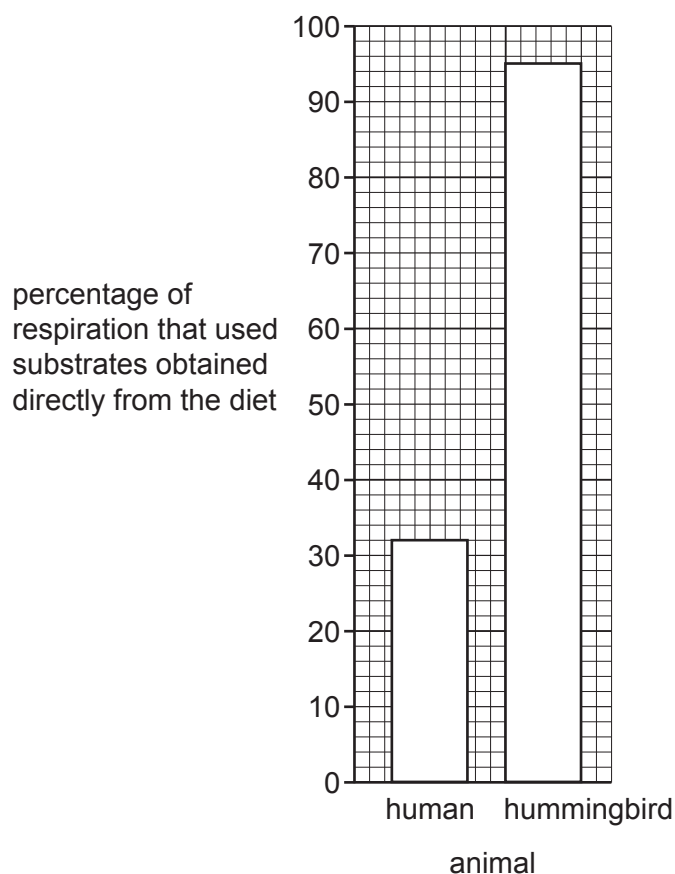


Fig. 6.3

Fig. 6.3 shows that hummingbirds have a higher percentage of respiration that uses substrates obtained directly from the diet.

Calculate how many times greater this figure is for hummingbirds compared to humans.

Show your working and write your answer to **two** decimal places.

answer = [2]

- (iv) Suggest the benefits to a hummingbird of obtaining most of its respiratory substrates from its diet while exercising.

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..... [2]

[Total: 15]